17. The marginal cost function of a commodity is given by $MC = \frac{14000}{\sqrt{7x+4}}$ and the fixed cost is ₹18,000. Find the total cost and average cost.

18. If the marginal cost (MC) of a production of the company is directly proportional to the number of units (x) produced, then find the total cost function, when the fixed cost is ₹ 5,000 and the cost of producing 50 units is ₹ 5,625.

19. If $MR = 20 - 5x + 3x^2$, find total revenue function.

20. If $MR = 14 - 6x + 9x^2$, find the demand function.

3.2.4 Consumer's surplus:

This theory was developed by the great economist Marshal. The demand function reveals the relationship between the quantities that the people would buy at a given price. It can be expressed as $p = f(x)$

Let us assume that the demand of the product $x = x_0$ when the price is p_0. But there can be some consumer who is ready to pay q_0 which is more than p_0 for the same quantity x_0. Any consumer who is ready to pay the price more than p_0 gains from the fact that the price is only p_0. This gain is called the consumer's surplus.

It is represented in the following diagram

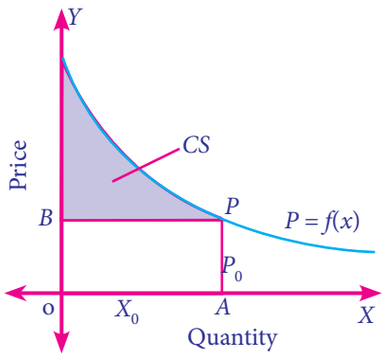


Fig. 3.16

Mathematically the Consumer's Surplus (CS) can be defined as

CS = (Area under the demand curve from $x = 0$ to $x = x_0$) – (Area of the rectangle OAPB)

$$CS = \int_0^{x_0} f(x)\,dx - x_0 p_0$$

Example 3.27

The demand function of a commodity is $y = 36 - x^2$. Find the consumer's surplus for $y_0 = 11$.

Solution:

Given $y = 36 - x^2$ and $y_0 = 11$

$11 = 36 - x^2$

$x^2 = 25$

$x = 5$

$$CS = \int_0^x (\text{demand function})\,dx - (\text{Price} \times \text{quantity demanded})$$

$$= \int_0^5 \left(36 - x^2\right) dx - 5 \times 11$$

$$= \left[36x - \frac{x^3}{3}\right]_0^5 - 55$$

$$= \left[36(5) - \frac{5^3}{3}\right] - 55$$

$$= 180 - \frac{125}{3} - 55 = \frac{250}{3}$$

Hence the consumer's surplus is $= \frac{250}{3}$ units.

3.2.5 Producer surplus

A supply function $g(x)$ represents the quantity that can be supplied at a price p. Let p_0 be the market price for the corresponding supply x_o. But there can be some producers who are willing to supply the commodity below the market price gain from the fact that the price is p_0. This gain is

XII Std - Business Maths & Stat EM Chapter 3.indd 73 2/28/2020 5:49:40 PM